Band-1 Computational Experiment: Odd Perfect Numbers — Lower Bounds & Factor-Chain Constraints

Human + ZoraASI full-dash (executor)

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Honesty stamp. This note does **not** claim that odd perfect numbers exist, and does **not** claim they do not exist. It is a **literature ledger** of lower bounds and necessary conditions, plus a **small on-box computational check** of classical filters (Euler form, Touchard congruences, abundancy) on synthetic candidates. Shrinking the search haystack is Band-1; settling existence is Band-4.

Abstract

Any odd perfect number N (if one exists) must satisfy a long list of necessary conditions. The standing published size bound is $N > 10^{1500}$ (Ochem–Rao 2012). We record the main ledger entries (ω, Ω, largest component, Euler form, Touchard) and run tiny classical checks: (i) Euclid–Euler even perfects verify $\sigma(N) = 2N$; (ii) exhaustive search finds **no** odd perfect in 1..50000; (iii) **312** synthetic Euler-form odd numbers all have $\sigma(N) \neq 2N$ (98 pass Euler+Touchard+mod-4 filters; closest abundancy ≈ 1.9845); (iv) factor-chain samples exhibit $\sigma(p^a)$-forced primes. **Existence status: OPEN.**

Problem statement

N is **perfect** if $\sigma(N) = 2N$, where σ is the sum-of-divisors function. Even perfect numbers are completely classified (Euclid–Euler: $N = 2^{k-1}(2^k - 1)$ with $2^k - 1$ Mersenne prime). Whether any **odd** perfect number exists is a classical open problem.

Literature ledger (not re-proved on box)

Constraint	Bound / form	Citation
Size	$N > 10^{1500}$	Ochem–Rao, *Math. Comp.* 81 (2012), 1869–1877 (DOI 10.1090/S0025-5718-2012-02563-4)
Prior size	$N > 10^{300}$	Brent–Cohen–te Riele, *Math. Comp.* 57 (1991)
Author-site push	$N > 10^{2200}$ (reported)	https://www.lirmm.fr/~ochem/opn/ — note separately from published 10^1500
Multiplicity	$\Omega(N) \geq 101$ (pub.); ≥ 115 (reported push)	Ochem–Rao 2012; author site

Constraint	Bound / form	Citation
Distinct primes	$\omega(N) \geq 9$; then ≥ 10	Nielsen *Math. Comp.* 76 (2007); Nielsen *Math. Comp.* (2015)
If $3 \nmid N$	$\omega(N) \geq 12$	Nielsen 2007
Largest component	$> 10^{62}$	Ochem–Rao 2012 (prior Cohen 1987: $> 10^{20}$)
Euler form	$N = \pi^{\alpha} M^2$, π prime, $\pi \equiv \alpha \equiv 1 \pmod 4$, $\gcd(\pi, M) = 1$	Euler (classical)
Touchard	$N \equiv 1 \pmod{12}$ or $N \equiv -1 \pmod{36}$	Touchard 1953

Method family (why Band-1). Factor-chain / branching arguments (Brent–Cohen–te Riele; Ochem–Rao): assume a small prime power $p^a \| N$, deduce that odd prime factors of $\sigma(p^a)$ divide N, branch on multiplicities, contradict either abundancy > 2 or N below the target size. This **forbids** regions of the haystack; it does not decide existence.

Method (box)

1. **Even-perfect sanity.** Rebuild Euclid–Euler perfects for Mersenne exponents $k = 2, 3, 5, 7$; assert $\sigma(N) = 2N$.
2. **Exhaustive odd scan.** For every odd $n \leq 50000$, compute $\sigma(n)$ and record perfects (expect none).
3. **Euler / Touchard / mod-4 filters.** On synthetic $N = \pi^{\alpha} q^2$ and $N = \pi^{\alpha} q^2 r^2$ with $\pi \equiv \alpha \equiv 1 \pmod 4$.
4. **Abundancy.** Measure $\sigma(N)/N$; count deficient vs abundant; report closest-to-2 among filter-passers.
5. **Factor-chain micro-demo.** For small odd p and exponents $a \in \{1, 2, 4, 6\}$, factor $\sigma(p^a)$ and list forced odd primes.
6. **Abundancy envelope (illustrative).** Crude $\prod p/(p-1)$ and a special+p^2 envelope — qualitative only; full OPN trees are far stricter because σ-factors force new primes with even exponents.

Results (numbers!)

Unit tests

`PASS`: even perfects $6, 28, 496, 8128$; 945 odd abundant not perfect; Euler accepts $5 \cdot 3^2$; Euler rejects 15.

Exhaustive odd search

- Limit: **50000**
- Odd integers checked: **25000**
- Odd perfects found: **0**
- Time: **0.081 s**

Synthetic Euler-form candidates

- Generated: **312** odd Euler-shaped numbers ($\pi \leq 40$, M-primes ≤ 25, $\alpha \in \{1, 5\}$)
- Euler-form OK: **312** (by construction)
- Touchard OK: counted in results; **mod-4 OK** automatic for Euler form
- Pass all classical filters tracked: **98**
- Perfect among them: **0**
- Deficient / abundant split: see `results.json` (majority deficient at this tiny scale)
- **Closest to abundancy 2:** $5^5 \cdot 3^2 \cdot 11^2$, abundancy $\approx$ **1.984492** ($\Delta \approx 0.0155$)
- All 98 filter-passers remain non-perfect: **True**

Factor-chain samples (illustrative)

Examples (full table in `results.json`):

Component	$\sigma(p^a)$	Forced odd primes (sample)
3^1	4	— (power of 2)
3^2	13	13
5^1	6	3
5^2	31	31
7^2	57	3, 19
127^1	128	—

This is exactly the first step of classical factor-chain haystack reduction: a hypothesized component forces new prime factors into N.

Key bound cited

Primary: $N > 10^{1500}$ (Ochem–Rao 2012).
Noted extension (author site): $N > 10^{2200}$, $\Omega(N) \geq 115$ — recorded but not elevated above the published theorem.

Why Band-1 vs Band-4

- **Band-1 (this note):** incremental constraints — size lower bounds, ω/Ω floors, Euler/Touchard filters, abundancy sieves, factor-chain forbidding of small prime powers. Each result **shrinks** the space of candidates.
- **Band-4:** a proof that an odd perfect number exists, or a proof that none exist. Still open; **not claimed**.

Reproducibility

- Code: `/workspace/experiments/band1/odd_perfect/`
- Run: `python3 odd_perfect_check.py`
- Results: `results.json`, `literature_ledger.json`, `SUMMARY.txt`
- Log: `/workspace/experiments/logs/2026-09-11_odd_perfect.log`

Conclusion

Literature: any odd perfect N satisfies $N > 10^{1500}$ (published), with much stronger structural constraints ($\omega \geq 10$, $\Omega \geq 101$, Euler form, ...). **Box:** classical filters and abundancy checks eliminate all **312** tiny synthetic Euler-form candidates; no odd perfect appears below $5 \cdot 10^4$. **Live Band-1 target:** continue haystack-shrinking / ledger hygiene — without claiming existence or non-existence.